Science doesn't have a knowledge problem. It has an infrastructure problem. The data exists, the talent exists, the questions exist — but they sit in disconnected silos, locked behind paywalls and misaligned incentives. This platform is the connective tissue.

01

What's Broken

Modern science runs on a publishing model designed in the 17th century. Researchers spend years producing a static PDF, submit it to anonymous gatekeepers, wait months for a verdict, and then — whether accepted or rejected — the underlying data, code, and failed experiments mostly disappear.

The result is staggering waste. Over **$28 billion annually** in the U.S. alone is estimated lost to irreproducible research. Negative results go unpublished. Duplication of effort is rampant. Brilliant researchers in underfunded institutions are invisible. And the incentive structure rewards novelty and volume over truth and rigor.

CURRENT MODEL	WHAT'S NEEDED
Static papers, frozen at publication	Living documents, continuously updated

Data buried in supplements or lost	Data, code, and methods inline and executable
Anonymous, one-shot peer review	Transparent, continuous community evaluation
Credit only for first authors on novel findings	Credit for every contribution — replication, review, curation
Funding decided by small committees	Diversified funding — grants, crowdfunding, microgrants
Knowledge stored as disconnected documents	A structured, queryable graph of claims and evidence

02

The Architecture

The platform is organized into seven interlocking layers. None is optional — they're designed to reinforce each other. Reproducibility tools feed the knowledge graph. The incentive system drives participation in peer review. The funding layer connects directly to open problem boards. Remove one layer and the system degrades.

◇ Layer 1 — Living Research Objects

Every project is a versioned, executable document — not a paper. It contains the narrative, the data, the code, the methods, and a full change history. Think of it as a Git repository that reads like a journal article.

Layer 2 — Reproducibility Engine

Containerized compute environments travel with every project. Anyone can re-run an analysis with one click. Automated checks flag statistical anomalies, undisclosed degrees of freedom, and missing dependencies. Every study gets a reproducibility score.

Layer 3 — Knowledge Graph

Claims are linked to evidence. Experiments are linked to hypotheses. Contradictions are surfaced, not hidden. Domain-specific ontologies interoperate through a shared schema layer, so connections across fields become visible. This is the map.

Layer 4 — Collaboration Infrastructure

Open problem boards publish unsolved questions. Skill-based matching connects statisticians with experimentalists, engineers with theorists. Micro-contribution workflows let anyone help — reviewing a dataset, suggesting a method, flagging an error — without joining a full project.

Layer 5 — Incentive & Reputation System

A contribution graph tracks every meaningful action: data curation, code review, replication, peer commentary, negative result reporting. Reputation accrues from verified accuracy over time, not publication count. This becomes a legible career record.

Layer 6 — Funding Layer

Embedded crowdfunding for specific experiments. Microgrants for small contributions. Transparent spend tracking. Pooled funding mechanisms — decentralized or institutional — connected directly to the problem boards. Good ideas shouldn't die for lack of a grant committee sponsor.

Layer 7 — AI Discovery Layer

Machine intelligence surfaces what humans miss: connections between distant fields, statistical inconsistencies, overlooked datasets, and hypothesis generation from structured knowledge. AI proposes, humans evaluate. The loop is always human-validated.

Design Principles

These are the non-negotiable commitments that distinguish this from another academic tool.

> **TRUTH OVER ATTENTION**
>
> The system must make careful, verified, replicated work more rewarding than flashy, unreproducible claims. Every incentive mechanism is tested against this standard.

> **PROCESS OVER PRODUCT**
>
> Science is not a paper. It's the ongoing cycle of question, experiment, analysis, revision, and challenge. The platform treats this cycle — not the snapshot — as the first-class object.

> **ACCESS WITHOUT COMPROMISE**
>
> Free or near-free for researchers everywhere. Multilingual. Lightweight enough for low-bandwidth environments. Independent researchers and small institutions are full citizens, not second-class users.

> **ACCOUNTABILITY WITHOUT GATEKEEPING**
>
> Verified identities for credibility. Pseudonymous contributions allowed with reputation tracking. Transparent history. The goal is trust through transparency, not exclusion through credentials.

SPEED WITHOUT RECKLESSNESS

Publication in minutes, not months. But every claim carries its evidence, its reproducibility score, its review history. Fast publishing paired with continuous, visible evaluation.

04

The Negative Results Archive

One of the highest-value, lowest-cost features: a dedicated, searchable registry of experiments that didn't produce the expected result — complete with methods, data, and analysis of why.

In pharma alone, millions of hours are wasted repeating experiments that someone else already knows don't work but never published. In materials science, in social psychology, in climate modeling — the pattern repeats everywhere. This archive isn't a graveyard. It's a map of explored territory that says **"we looked here — look elsewhere."**

THE MATH

If even 10% of failed experiments in biomedicine were searchable, conservative estimates suggest it would save 3–5 years of redundant work per research domain per decade. For a platform feature that's essentially "let people upload what didn't work," the return on investment is extraordinary.

05

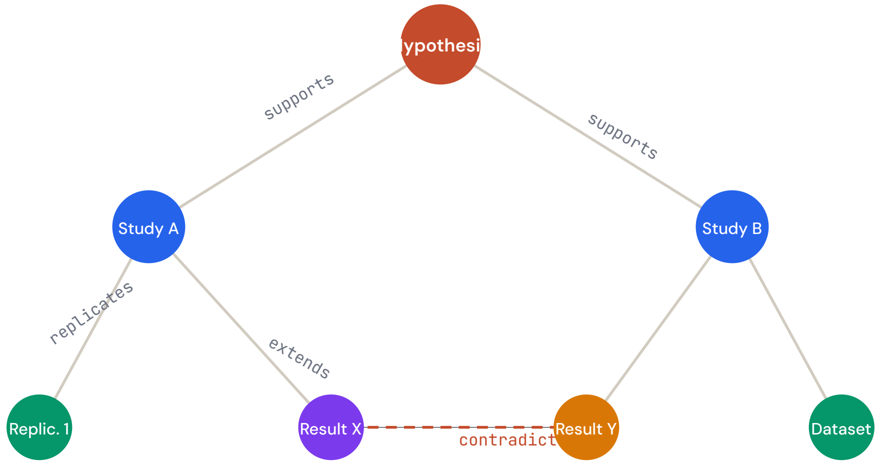
The Knowledge Graph — In Practice

This is the most technically ambitious layer, and the one that separates a useful tool from a transformative platform.

A universal ontology across all of science is a fool's errand. Biology, physics, and social science don't share vocabulary, units, or epistemological standards. The approach instead is **federated domain schemas** — each field maintains its own structured vocabulary, with a thin interoperability layer that maps shared concepts (causation, measurement, uncertainty, replication) across domains.

Every claim on the platform is a node. Every piece of evidence is a node. The edges between them are typed: *supports, contradicts, replicates, extends, depends on*. When a researcher opens any question, they see not a list of papers but a living map — where the evidence clusters, where it conflicts, and where the blank spaces are.

SIMPLIFIED KNOWLEDGE SUBGRAPH



06

The Flywheel

The layers aren't a checklist — they're a feedback loop. Each one makes the others more valuable.

Open problem boards attract collaborators. Collaborators produce reproducible research objects. Reproducible work feeds the knowledge graph. The knowledge graph reveals gaps. Gaps attract funding. Funding enables new work on the problem boards. The AI layer accelerates every step by surfacing connections, flagging errors, and suggesting directions.

This is the critical insight: **previous platforms digitized one part of the old system**. arXiv digitized distribution. GitHub digitized code sharing. PubMed digitized search. Each is valuable in isolation. But none of them created a flywheel — because a flywheel requires the layers to talk to each other.

THE ANALOGY

Think of it this way. arXiv is a library. GitHub is a workshop. PubMed is a card catalog. Slack is a hallway. Kickstarter is a tip jar. The Open Science Engine is the university that contains all of them — with shared hallways, a common directory, and a single reputation that follows you everywhere.

07

What Success Looks Like

In three years, a postdoc in Nairobi spots a gap on the knowledge graph in protein misfolding, proposes a hypothesis, receives a microgrant funded by a pooled mechanism, recruits a statistician from São Paulo through skill matching, runs her analysis on the platform's containerized compute, publishes a living research object with inline code and data, gets immediate community feedback, and watches as three labs replicate her work within months — all without a single journal submission.

Her contribution graph shows the full picture: not just the final result, but the data she curated for two other projects, the peer reviews she wrote, the negative result she reported that saved someone else a year of work. That graph is what her institution looks at for tenure.

That's not a utopian fantasy. Every piece of that technology exists today. What doesn't exist is the single platform that stitches them together with a coherent incentive model.

That's what we build.

"The best way to predict the future of science is to build its infrastructure."